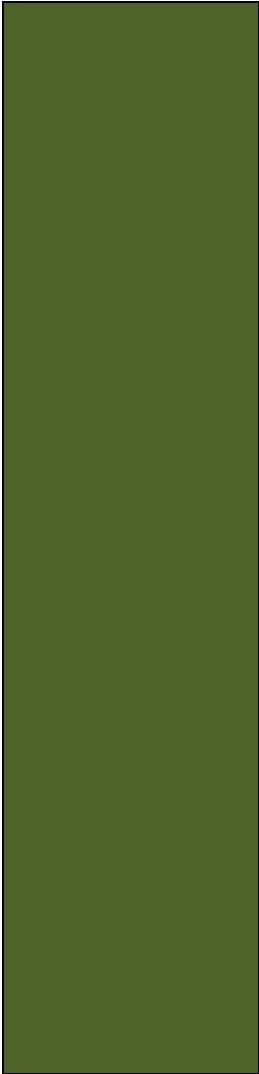


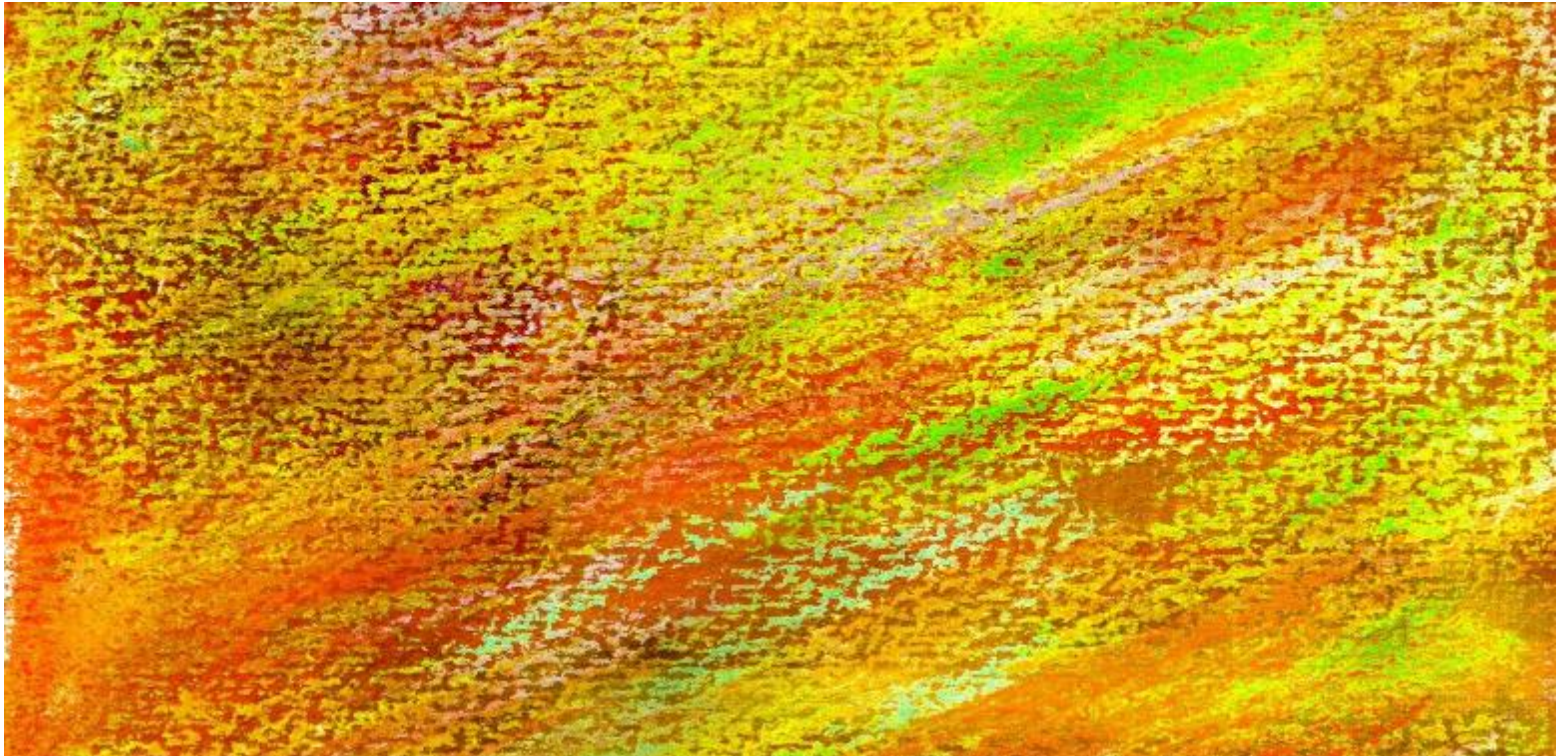


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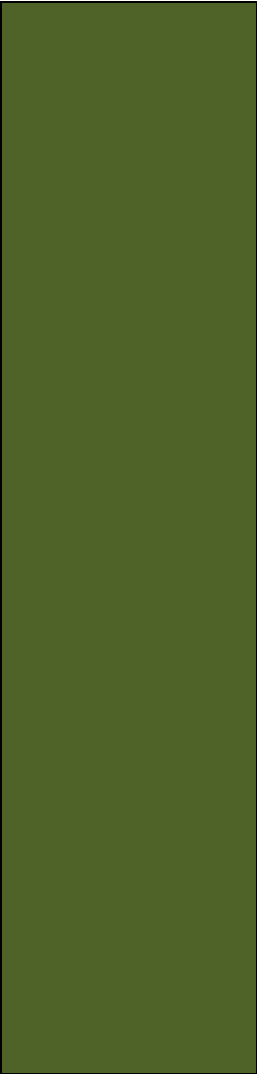
Chapter Two

The Relational Model





Chapter Objectives

- 
- Learn the conceptual foundation of the relational model
 - Understand how relations differ from nonrelational tables
 - Learn basic relational terminology
 - Learn the meaning and importance of keys, foreign keys, and related terminology
 - Understand how foreign keys represent relationships



Relational Data Structure

- **Relation** – a relation is a table with columns and rows
- **Attribute** – an attribute is a named column and rows
- **Domain** – a domain is the set of allowable values for one or more attributes
- **Tuple** – a tuple is a row of a relation
- **Degree** – the degree of a relation is the number of attributes it contains
- **Cardinality** – the cardinality of a relation is the number of tuples it contains
- **Relational Database** - collection of normalized relations with distinct relation names.

Example of a Relation

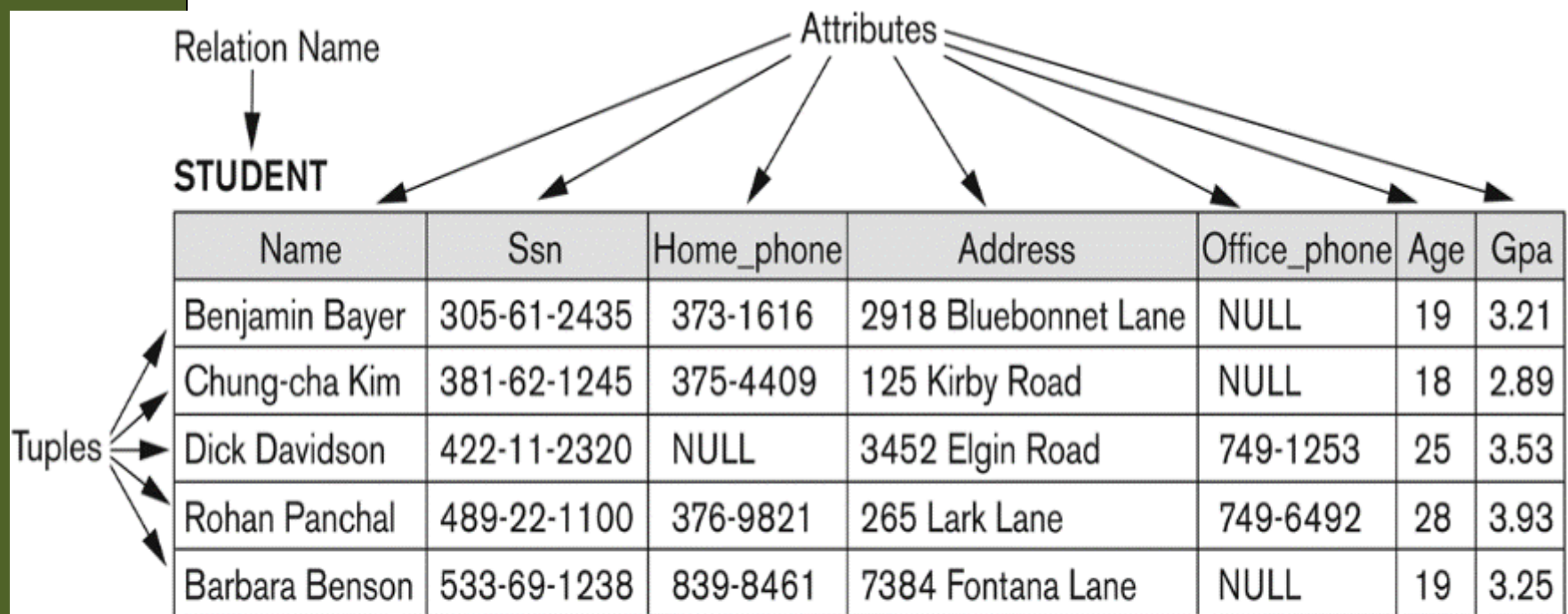


Figure 5.1

The attributes and tuples of a relation STUDENT.



Characteristics of a Relation

1. Rows contain data about an entity.
2. Columns contain data about attributes of the entity.
3. Cells of the table hold a single value.
4. All entries in a column are of the same kind.
5. Each column has a unique name.
6. The order of the columns is unimportant.
7. The order of the rows is unimportant.
8. No two rows may be identical.

EmployeeNumber	FirstName	LastName
100	Mary	Abernathy
101	Jerry	Cadley
104	Alex	Copley
107	Megan	Jackson



A Nonrelation Example

Cells of the table hold multiple values

EmployeeNumber	Phone	LastName
100	335-6421, 454-9744	Abernathy
101	215-7789	Cadley
104	610-9850	Copley
107	299-9090	Jackson



Example of a Nonrelational Table

No two rows may be identical



EmployeeNumber	Phone	LastName
100	335-6421	Abernathy
101	215-7789	Cadley
104	610-9850	Copley
100	335-6421	Abernathy
107	299-9090	Jackson



Terminology

Synonyms...

Table	Row	Column
File	Record	Field
Relation	Tuple	Attribute

Figure 2-6: Equivalent Sets of Terms



Composite Attributes

Composite
Attributes

name

first-name middle-initial last-name

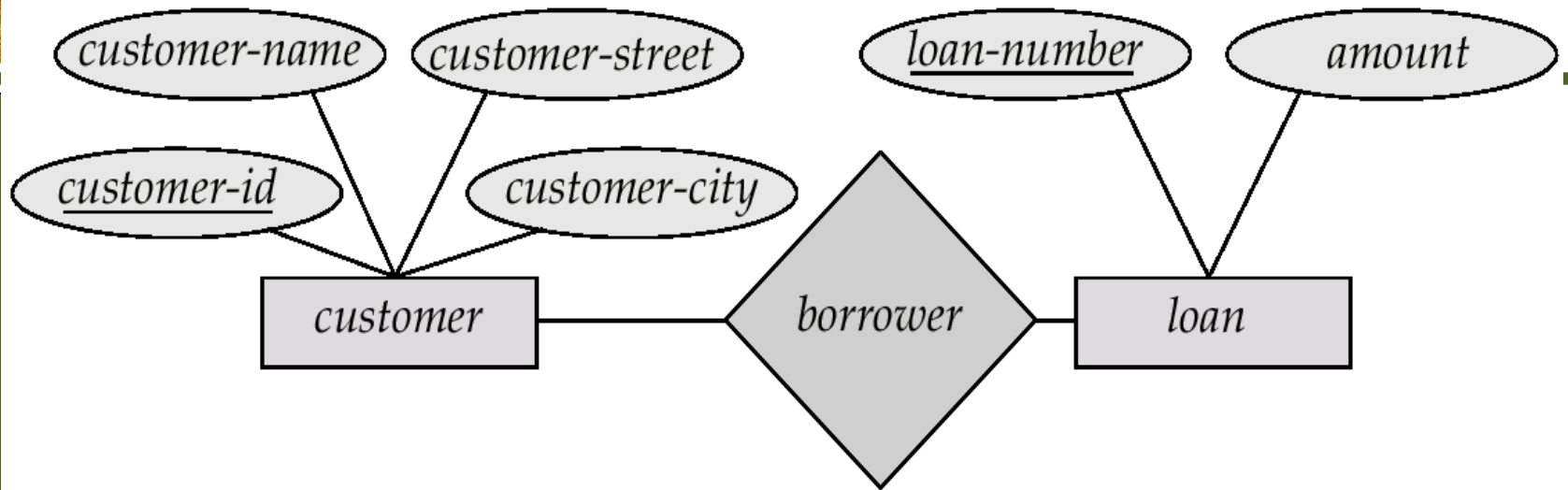
Component
Attributes

address

street city state postal-code

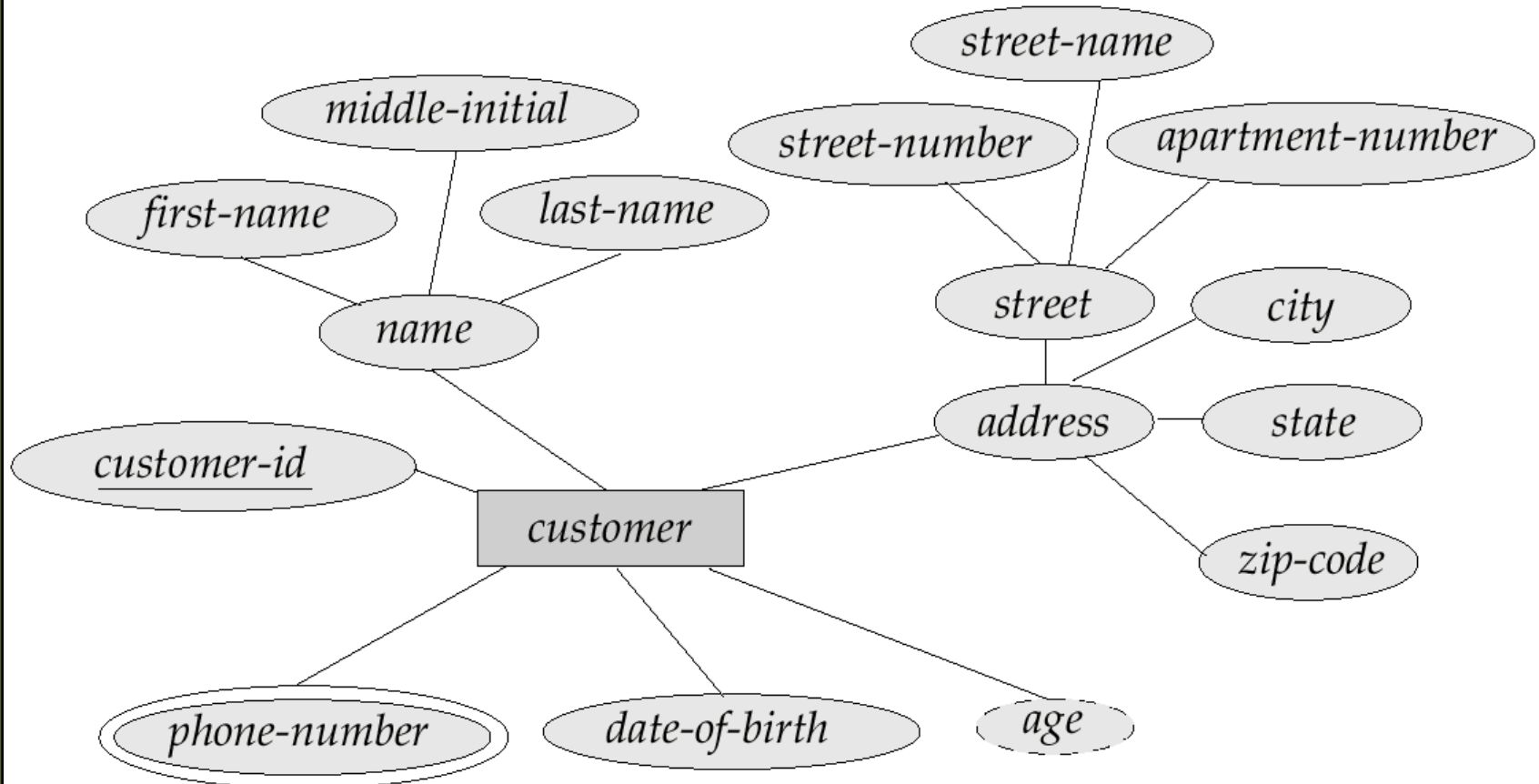
street-number street-name apartment-number

E-R Diagrams

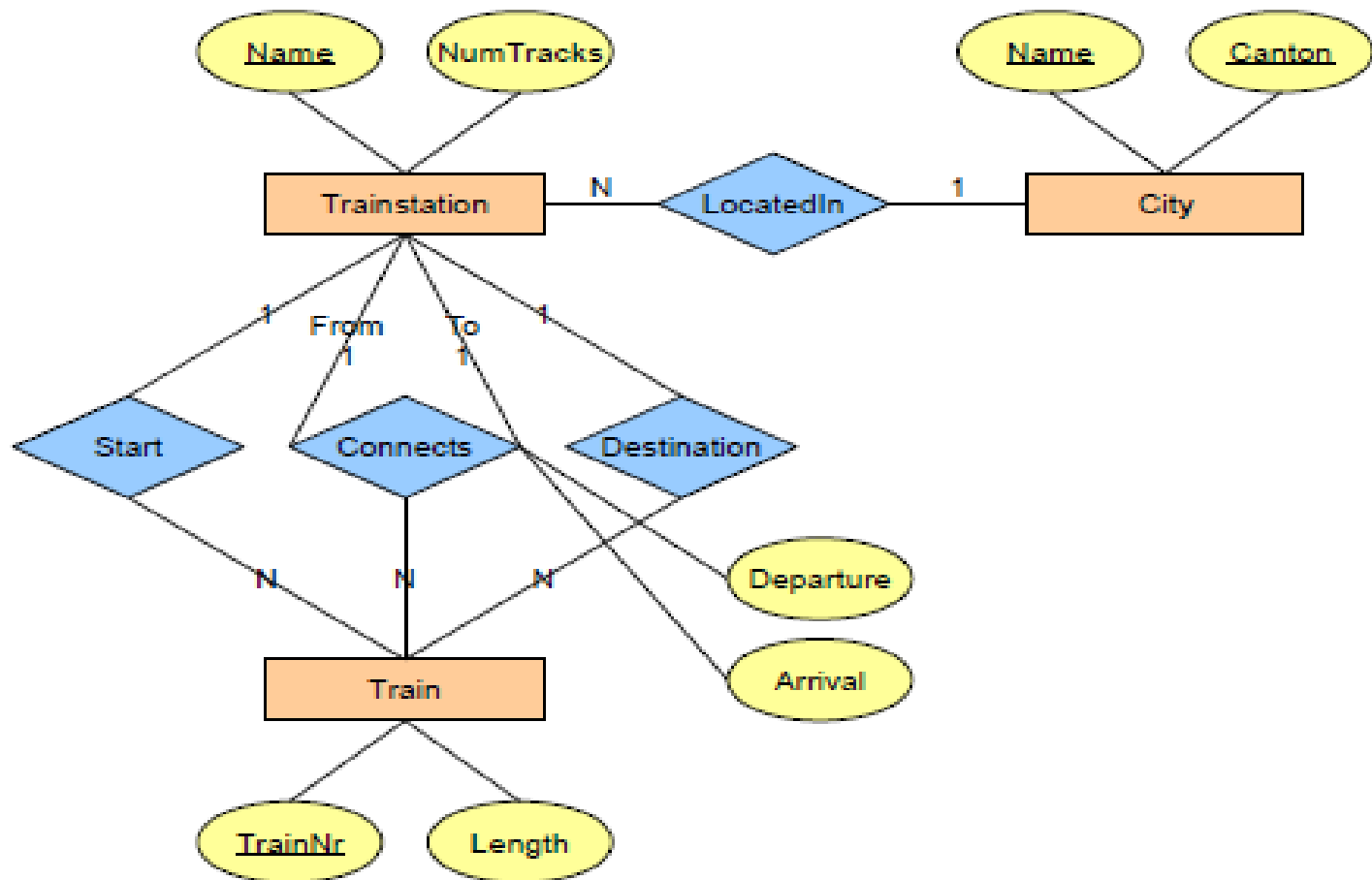


- a) **Rectangles** represent entity sets.
- b) **Diamonds** represent relationship sets.
- c) **Lines** link attributes to entity sets and entity sets to relationship sets.
- d) **Ellipses** represent attributes
 - a) **Double ellipses** represent multivalued attributes.
 - b) **Dashed ellipses** denote derived attributes.
- e) **Underline** indicates primary key attributes (will study later)

E-R Diagram With Composite, Multivalued, and Derived Attributes

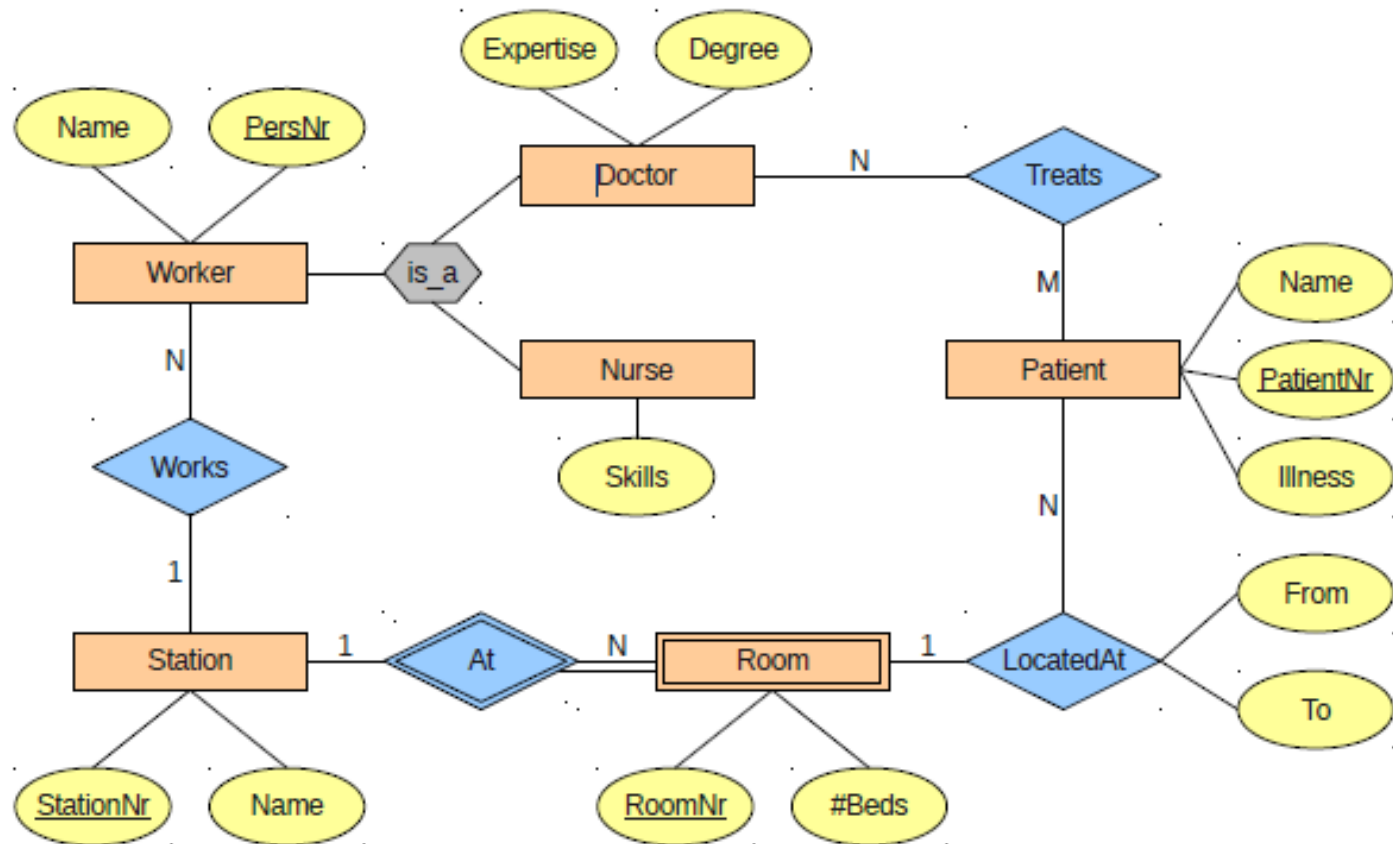


E-R Diagram With Composite, Multivalued, and Derived Attributes



E-R Diagram With Composite, Multivalued, and Derived Attributes

Hospital





A Key

- A **key** is one (or more) column(s) of a relation that is (are) used to identify a row.



Uniqueness of Keys

Unique Key	Nonunique Key
Data value is unique for each row. Consequently, the key will uniquely identify a row.	Data value may be shared among several rows. Consequently, the key will identify a set of rows.



A Composite Key

- A **composite key** is a key that contains two or more attributes.
- For a key to be unique, it must often become a composite key.



Composite Key Characteristics

- To identify a family member, you need to know a FamilyID, a FirstName, and a Suffix (e.g., Jr.).
- The composite key is:
(FamilyID, FirstName, Suffix).
- One needs to know the value of all three columns to uniquely identify an individual.



A Candidate Key

- A **candidate key** is called “candidate” because it is a candidate to become the primary key.
- A candidate key is a unique key.



A Primary Key

- A **primary key** is a candidate key chosen to be the main key for the relation.
- If you know the value of the primary key, you will be able to uniquely identify a single row.



Relationships Between Tables

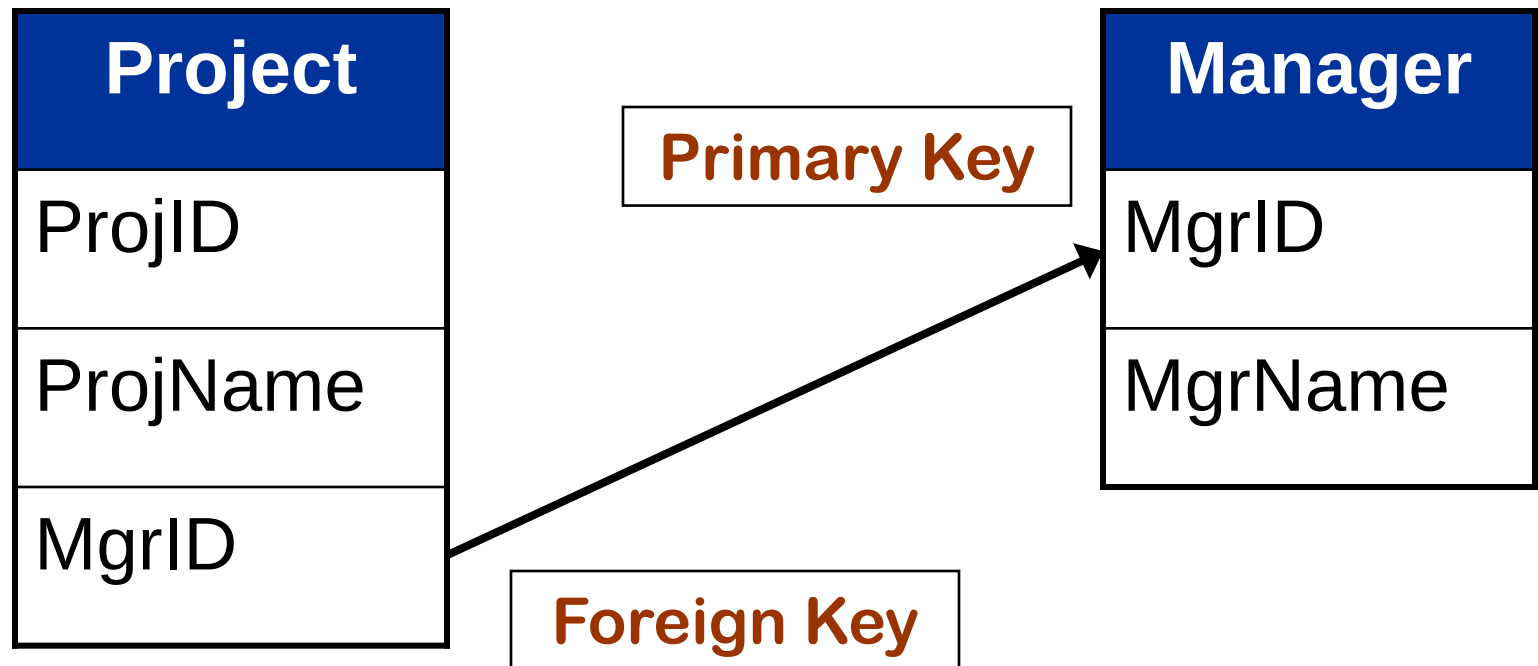
- A table may be related to other tables.
- For example
 - An Employee works in a Department
 - A Manager controls a Project



A Foreign Key

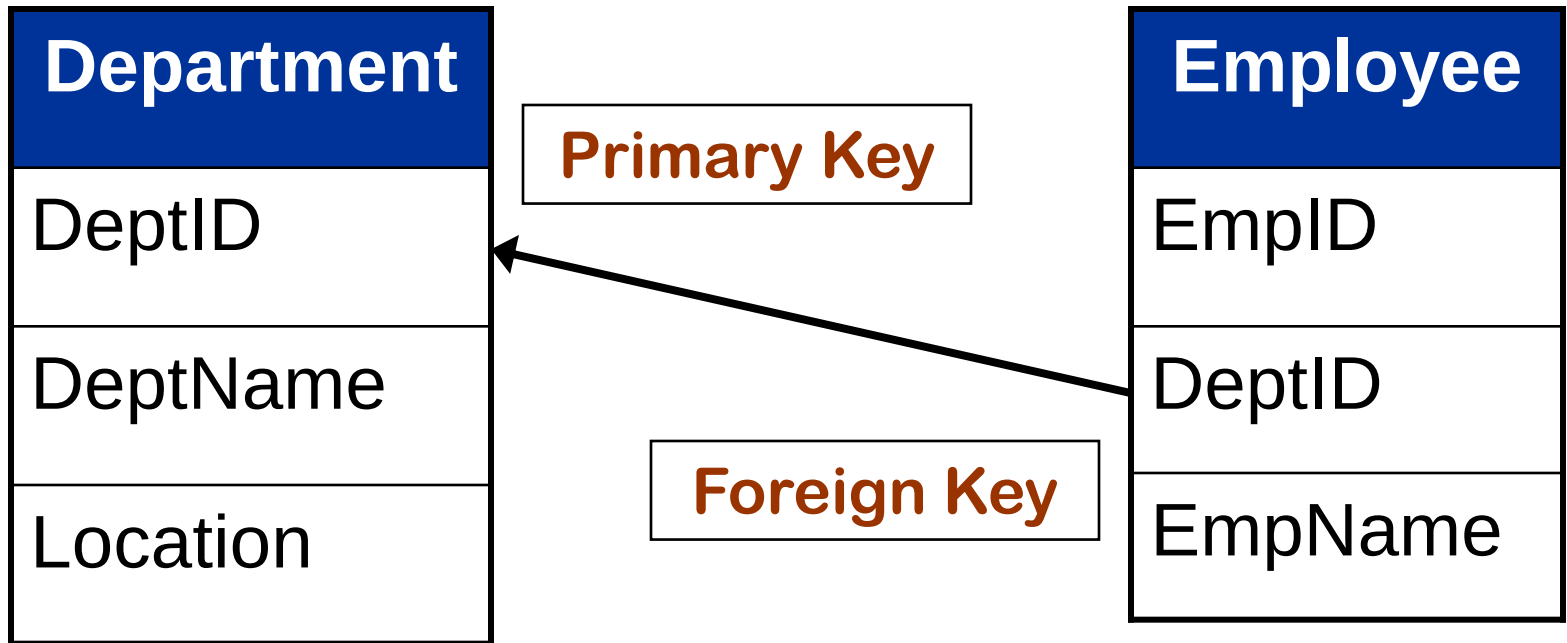
- To preserve relationships, you may need to create a **foreign key**.
- A foreign key is a primary key from one table placed into another table.
- The key is called a foreign key in the table that received the key.

Foreign Key Example I



Foreign Key Example II

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Referential Integrity

- **Referential integrity** states that every value of a foreign key must match a value of an existing primary key.
- Example (see previous slide):
 - If EmpID = 4 in EMPLOYEE has a DeptID = 7 (a foreign key), a Department with DeptID = 7 must exist in DEPARTMENT.
 - The primary key value must exist before the foreign key value is entered.



Referential Integrity (Cont'd)

- Another perspective...

*The value of the Foreign Key EmployeeID
in EQUIPMENT*

must exist in

*The values of the Primary Key EmployeeID
in EMPLOYEE*



The Null Value

- A **Null value** means that no data was entered.
- This is different from a zero, space character, or tab character.



The Problem of Null Values

- A Null is often ambiguous. It could mean...
 - The column value is not appropriate for the specific row.
 - The column value is not decided.
 - The column value is unknown.
- Each may have entirely different implications.



Functional Dependency

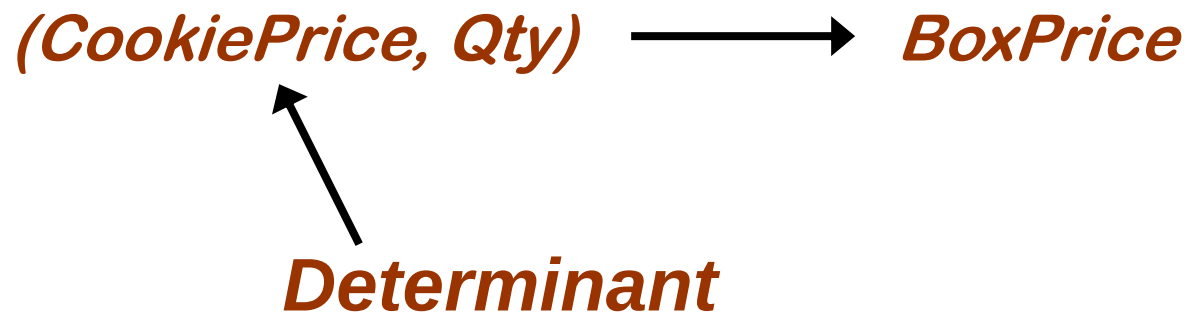
- Functional Dependency—A relationship between attributes in which one attribute (or group of attributes) determines the value of another attribute in the same table
- Illustration...
 - The price of one cookie can determine the price of a box of 12 cookies.

(CookiePrice, Qty) → BoxPrice



Determinants

- The attribute (or attributes) that we use as the starting point (the variable on the left side of the equation) is called a *determinant*.





Candidate/Primary Keys and Functional Dependency

- By definition...
A candidate key of a relation will functionally determine all other attributes in the row.
- Likewise, by definition...
A primary key of a relation will functionally determine all other attributes in the row.



Primary Key and Functional Dependency Example

(EmployeeID) → *(EmpLastName,
EmpPhone)*

(ProjectID) → *(ProjectName,
StartDate)*



END OF CHAPTER 2